CONSISTENCY OF MARTIN AXIOM WITH A MAXIMAL PAIRWISE-FAR COUNTRYMAN FAMILY

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Allowed standard prerequisites (editorial).

- Coherent Aronszajn trees, Countryman lines, diamond and ccc iteration. Native club-isomorphism and far-line definitions, suitable-tree Xi/Xi+ invariants, safe forcing, uniform coherent extension tools and all cited prerequisites retained.. Complete nonadjacent author proof chain included; no extra count for safe forcing, diamond existence or consequent far-line statement.

1. Introduction

A result of Shelah [13] shows that there is an uncountable linear order C such that its square $C \times C$ is the union of countably many chain with the coordinate-wise partial order. Such linear orders are called *Countryman lines*, and are necessarily Aronszajn, i.e., contain no copies of uncountable real orders, ω_1, or its reverse $\omega_1^\star$. Also, if C is any Countryman line, then C and $C^\star$ are *far*, i.e., no uncountable linear order embeds into both.

Recall that a basis for a class of linear orders is a set $\mathcal{B}$ such that any linear order in the class contains an isomorphic copy of an order in $\mathcal{B}$. It follows from the previous paragraph that any basis for the Countryman lines, and thus for the Aronszajn lines also, must contain at least two elements. Shelah [13] conjectured that it is consistent that any Aronszajn line contains a Countryman suborder. This was finally proved by Moore [6]; PFA implies that if C is any Countryman line, then C and $C^\star$ form a basis for the Aronszajn lines, that is, any Aronszajn line contains a suborder isomorphic to C, or to $C^\star$.

Observe that Moore's theorem implies that any Countryman line is a minimal uncountable linear order, and that if C and C' are Countryman lines, then they are near or reverse near, i.e, there is an uncountable suborder of C which is isomorphic or reverse-isomorphic to a suborder of C'. These two assumptions together imply

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that if C is any Countryman line, then C and $C^\star$ form a basis for the Countryman lines. It was already known to Abraham-Shelah [1] that these statements have no large cardinal strength, a fact which is still not known for Moore's theorem.

In [15, 2.1.12] (see [7] for a more detailed proof), Todorcevic proved that already MA_{ω_1} implies that any Countryman line is minimal, and at some point, the earliest I know of is [15, 2.1.13], it was also claimed that MA_{ω_1} implies that any two Countryman lines are either equivalent (bi-embeddable) or reverse-equivalent. However, it appears that several people[1] have realized that MA_{ω_1} seems to not be enough for this, but that MA_{ω_1} + "Any two normal Aronszajn trees are club-isomorphic" is[2]. The later statement has no large cardinal strength either, and follows from PFA restricted to posets of size ω_1. The seemingly incorrect claim was cited by several authors and has led to some confusion. In particular, the article of the author [11] makes this mistake[3], as well as [7, Theorem 1.1]. It should be noted that the proofs in [7, 11] are correct, but the conclusion drawn are stronger due to the assumption of the claim. See also [8, Remark 4.4].

The objective of this paper is to make this precise, and prove that in fact MA_{ω_1} is consistent with a family of 2^{ω_1}-many pairwise far Countryman lines, the maximum possible value. In doing this, we will look closely to the problem of club-isomorphisms between coherent Aronszajn trees. For this we will apply the non-dangerous forcing method of Abraham-Shelah [1, Section 6]. However, there will be subtle differences that will require to re-do most of the proofs.

Our main theorem is the following.

Theorem A. *Consistently with $\mathsf{MA} + 2^\omega = \omega_2$ there is a family of 2^{ω_1}-many coherent Aronszajn trees which pairwise contain no club-isomorphic subtrees. Consequently, $\mathsf{MA} + 2^\omega = \omega_2$ is consistent with the existence of 2^{ω_1}-many Countryman lines which are pairwise far.*

Terminology and notation: Our notation on trees is mostly standard. For a tree T, $t \in T$, and $\xi \leq \mathrm{ht}(t)$, we let $t \restriction \xi$ be the unique $t' \leq_T t$ such that $t' \in T_\xi$ (the ξ-th level of T). For $t, s \in T$ we let $t \perp s$ mean that t and s are incomparable, $\Delta(t,s)$ denotes $\sup\{\xi \leq \mathrm{ht}(t) \cap \mathrm{ht}(s) : t \restriction \xi = s \restriction \xi\}$. If T is a tree of height ω_1, by a *subtree* of T, we mean a downwards closed uncountable subset of T. If $X \subseteq \omega_1$, we let $T \restriction X = \bigcup_{\alpha \in X} T_\alpha$. Given trees $\{T_i : i \in I\}$, $\bigotimes_{i\in I} T_i$ is the partial order given by the coordinate-wise order on the set of $a \in \prod_{i\in I} T_i$ satisfying $\mathrm{ht}(a(i)) = \mathrm{ht}(a(j))$ for all $i, j \in I$. This is again a tree. We will mainly work with subtrees of $\omega^{<\omega_1}$, where the order is sequence extension. A subtree T of $\omega^{<\omega_1}$ is called *coherent* if for every $t, s \in T$ satisfying $\mathrm{dom}(t) \leq \mathrm{dom}(s)$, $t =^* s \restriction \mathrm{dom}(t)$. Here, $f =^* g$ denotes that f and g are functions with the same domain that differ on finitely many values. T is called *finite-to-one* if every $t \in T$ is a finite-to-one function.

Let T and S be ω_1-trees. We say that T *club-embeds* into S if there is club C and a tree embedding $f : T \restriction C \to S$. By standard arguments we may assume that f is height preserving, so that in particular $\mathrm{ran}(f) \subseteq S \restriction C$, f is called a *club-isomorphism* if $\mathrm{ran}(f) = S \restriction C$. Observe that T and S have club-isomorphic subtrees if and only if some subtree of T club-embeds into S.

For a closed $C \subseteq \omega_1$ and $\xi \in \omega_1$, we let $\nu_C(\xi)$ be the greatest $\nu \in C \cup \{0\}$ such that $\nu \leq \xi$.

[1] In particular, this was pointed out to me during a talk by a mathematician which unfortunately I was unable to identify.

[2] The gap in the proof of [15, 2.1.13] is easily overcome using the extra assumption.

[3] Fortunately, the only result of that article that suffered from this is Lemma 7.1, in which the hypothesis of MA_{ω_1} must be complemented with club-isomorphisms.

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2. Preliminary analysis

Recall that our main objective is to construct models of MA_{ω_1} with large Countryman line basis. We first explain how to reduce this problem to club-isomorphisms between coherent trees. We will work with a special but common type of coherent trees, those without Suslin subtrees. Moreover we want this property to be preserved by any ω_1-preserving forcing extensions. For this we adopt the following convention: we will say that T is *suitable* if it is a coherent subtree of $\omega^{<\omega_1}$ of height ω_1, which is either special or finite-to-one. Such trees have the following easy properties which we will be using frequently.

Fact 2.1. *Let T be suitable. Then T is an Aronszajn tree, every uncountable subset of T contains an uncountable antichain, and T remains suitable in any ω_1-preserving forcing extension.*

The following lemmas make the connection between coherent trees, embeddings between Countryman lines, and club-isomorphisms.

Fact 2.2. *Let T be suitable. The natural lexicographic ordering of T, $(T, <_{\mathrm{lex}})$, is a Countryman line.*

For a proof of the finite-to-one case see [16, Theorem 2.5], and for the special case see [15, Theorem 4.4.1]. Also [10, Corollary 1] handles both at the same time.

Finally the following makes the connection with club-isomorphisms, the proof is standard but somewhat tedious, a written proof (for the increasing case) appears in [12, Lemma 5.2], which in turns cites [9].

Fact 2.3. *Suppose that T and S are lexicographically ordered Hausdorff Aronszajn tree. If there is a $<_{\mathrm{lex}}$-increasing or $<_{\mathrm{lex}}$-decreasing map from an uncountable subset of T into S, then some subtree of T club-embeds into S.*

From this we conclude the following.

Proposition 2.4. *If $\mathcal{T}$ is a family of suitable trees such that for all distinct $T, S \in \mathcal{T}$, T and S have no club-isomorphic subtrees, then $\{(T, <_{\mathrm{lex}}), (T, <^*_{\mathrm{lex}}) : T \in \mathcal{T}\}$ is a family of pairwise far Countryman lines, and thus any basis for the Countryman lines has size at least $2|\mathcal{T}|$.*

Our task then, is to construct models of MA_{ω_1} where there are large families as above. For this we need two things, to be able to construct suitable trees without club-isomorphic subtrees, and to preserve this.

In the model of [1], they iterate ccc forcings while preserving a property on a single Aronszajn trees T, which implies that no two distinct cones of T have club-isomorphic subtrees. We cannot do the same thing and expect T to be coherent; MA_{ω_1} implies that such trees are club minimal, in the sense that they club-embed into all its subtrees [14].

Fix T and S suitable trees. It is not hard to see that T and S have no club-isomorphic subtrees iff the following holds: for every club C and uncountable $A \subseteq T \otimes S$, there are $a, b \in A$ such that $a(0) \perp b(0)$, $a(1) \perp b(1)$ and $\nu_C(\Delta(a(0), b(0))) \neq \nu_C(\Delta(a(1), b(1)))$. Imitating [1], we could work with just this idea to obtain a model

of MA_{ω_1} + "there are suitable trees T and S without club-isomorphic subtrees", thus obtaining a model of MA_{ω_1} in which no basis for the Countryman lines has less than 4 elements. However, we will need a slightly more general version to increase the cardinality of the basis

Let $T^0, \dots, T^{n-1}$ be Hausdorff ω_1 trees. For a club C and $a, b \in T^0 \otimes \cdots \otimes T^{n-1}$, we say that a and b *conflict on* C (or are conflicting on C) if for every $i < n$, $a(i) \perp b(i)$, and for every $i < j < n$, and $\nu_C(\Delta(a(i), b(i))) \neq \nu_C(\Delta(a(j), b(j)))$. We will work with the following statements which generalize our previous discussion:

$\Xi(T^0, \dots, T^{n-1})$ For every club C, and for every uncountable $A \subseteq \otimes_{i<n} T^i$, there are $a, b \in A$ conflicting on C.

$\Xi^+(T^0, \dots, T^{n-1})$ For every club C, and for every uncountable $A \subseteq \otimes_{i<n} T^i$, there is an uncountable $B \subseteq A$ such that the elements of B pairwise conflict on C.

Definition 2.5. Let $\mathcal{T}$ be a family of ω_1-trees. By $\Xi(\mathcal{T})$ we mean that $\Xi(T_0, \dots, T_{n-1})$ holds whenever $T_0, \dots, T_{n-1} \in \mathcal{T}$ are distinct. $\Xi^+(\mathcal{T})$ is defined analogously.

If $\Xi(\mathcal{T})$ holds, then any two trees in $\mathcal{T}$ have no club-isomorphic subtrees, and thus by Theorem 2.4, if $\mathcal{T}$ consists of suitable trees, any basis for the Countryman lines has size at least $2|\mathcal{T}|$. It is clear that $\Xi(\mathcal{T})$ is formally stronger than "the elements of $\mathcal{T}$ pairwise contain no club-isomorphic subtrees". This will be important for proving the ccc of certain posets.

We have two tasks, to obtain models satisfying $\Xi(\mathcal{T})$ for a family of suitable trees with cardinality as large as possible, and then to iterate ccc forcings preserving Ξ.

3. Non-dangerous forcings

This section closely follows Abraham-Shelah's [1, Section 6], so familiarity with it will make it easier to follow.

For this section fix a nonempty family $\mathcal{T}$ of suitable trees. As a convention, we will let $\mathcal{A}(\mathcal{T})$ denote the set of A such that $A \subseteq \bigotimes_{i<n} T^i$ for some $1 \leq n < \omega$ and distinct $T^0, \dots, T^{n-1} \in \mathcal{T}$.

Definition 3.1. A forcing poset $\mathbb{P}$ is called $\mathcal{T}$*-dangerous* if for some club C and binary relation R such that $\mathrm{dom}(R) \subseteq \mathbb{P}$ and $\mathrm{ran}(R)$ is an uncountable member of $\mathcal{A}(\mathcal{T})$, whenever $(p, a), (p', a') \in R$ and p is compatible with p', a does not conflict with a' on C. We say that $\mathbb{P}$ is $\mathcal{T}$-safe if it is not $\mathcal{T}$-dangerous.

Observe that $\mathcal{T}$-safe implies being ccc.

Lemma 3.2. *Assume $\Xi(\mathcal{T})$. If $\mathbb{P}$ is any $\mathcal{T}$-safe forcing, then $\mathbb{P} \Vdash \Xi(\mathcal{T})$.*

Proof. Suppose the conclusion fails. We will show that $\mathbb{P}$ is not $\mathcal{T}$-safe. Since $\mathbb{P} \nVdash \Xi(\mathcal{T})$, for some $p \in \mathbb{P}$, names $\mathring{A}$ and $\mathring{C}$, and distinct $T_0, \dots, T_{n-1} \in \mathcal{T}$, we have that p forces $\mathring{A} \subseteq T_0 \otimes \cdots \otimes T_{n-1}$ is uncountable, $\mathring{C}$ is a club, and that no two elements of $\mathring{A}$ conflict on $\mathring{C}$. Since $\mathbb{P}$ is ccc, we may assume that $\mathring{C}$ is a club in the ground model, call it C (conflicting in a coarser club implies conflicting in the finer).

For each $\xi < \omega_1$, one finds $p_\xi \leq p$ and $a_\xi \in T_0 \otimes \cdots \otimes T_{n-1}$ of height no less than ξ, such that $p_\xi \Vdash a_\xi \in \mathring{A}$. Let $R := \{(p_\xi, a_\xi) : \xi < \omega_1\}$. It should be clear that $\mathrm{ran}(R)$ is an uncountable subset of $T_0 \otimes \cdots \otimes T_{n-1}$, and that whenever p_ξ and p_η are compatible, a_ξ and a_η do not conflict on C. So R witnesses that $\mathbb{P}$ is not $\mathcal{T}$-safe. □

Remark 3.3. Note that if we assume that $\mathbb{P}$ is ccc, then the converse also holds.

The following is clear, but important.

Lemma 3.4. *Assume $\Xi^+(\mathcal{T})$. Every ccc forcing is $\mathcal{T}$-safe.*

The following says that we can iterate $\mathcal{T}$-safe posets. The argument is very similar to the classical proof that the ccc is preserved under finite support iterations. The analogous in [1] is given without proof so we give a sketch.

Lemma 3.5. *Assume $\Xi(\mathcal{T})$. Being $\mathcal{T}$-safe is preserved under finite support iterations.*

Proof. The proof is by induction on the length of the iteration. The limit case follows from the induction hypothesis exactly as in the proof for the preservation of the ccc. We prove the successor step.

Suppose $\mathbb{P}$ is $\mathcal{T}$-safe and that $\mathring{\mathbb{Q}}$ is a $\mathbb{P}$-name such that $\mathbb{P} \Vdash$ "$\mathring{\mathbb{Q}}$ is $\mathcal{T}$-safe". Let C be a club, and R be such that $\mathrm{dom}(R) \subseteq \mathbb{P} * \mathring{\mathbb{Q}}$ and $\mathrm{ran}(R)$ is an uncountable member of $\mathcal{A}(\mathcal{T})$. Let $\langle((p_\xi, \mathring{q}_\xi), a_\xi) : \xi < \omega_1\rangle$ enumerate a subset of R satisfying that the a_ξ are all distinct. We need to show that there are $\xi, \eta < \omega_1$ such that $(p_\xi, \mathring{q}_\xi)$ and $(p_\eta, \mathring{q}_\eta)$ are compatible, and a_ξ and a_η are conflicting on C.

Let $\mathring{\Gamma} := \{(\check{\xi}, p_\xi) : \xi < \omega_1\}$. Then the fact that $\mathbb{P}$ is ccc implies that $\mathbb{P} \nVdash$ "$\mathring{\Gamma}$ is countable". Thus for some generic G for $\mathbb{P}$, $\Gamma := \mathring{\Gamma}[G]$ is uncountable. Since $\mathbb{P} \Vdash$ "$\mathring{\mathbb{Q}}$ is $\mathcal{T}$-safe", there are $\xi \neq \eta$ in Γ such that $\mathring{q}_\xi[G] \not\perp \mathring{q}_\eta[G]$ and a_ξ conflicts with a_η in C. Thus for some $p' \in G$ and $\mathring{q}'$, $p' \Vdash \mathring{q}' \leq \mathring{q}_\xi, \mathring{q}_\eta$. Since p_ξ, p_η and p' are in G, we may assume that $p' \leq p_\xi, p_\eta$. Thus $(p', \mathring{q}')$ witnesses that $(p_\xi, \mathring{q}_\xi)$ and $(p_\eta, \mathring{q}_\eta)$ are compatible. □

Remark 3.6. Observe that in the previous proof we do not use that $\mathbb{P}$ is $\mathcal{T}$-safe, only that $\mathbb{P}$ is ccc. However its use is hidden: if $\mathbb{P}$ is not $\mathcal{T}$-safe, then $\mathbb{P} \nVdash \Xi(\mathcal{T})$ by Theorem 3.3, so $\mathbb{P} \nVdash$ "$\mathring{\mathbb{Q}}$ is $\mathcal{T}$-safe".

We now want show that given a family $\mathcal{T}$ of suitable trees satisfying $\Xi(\mathcal{T})$, a club C, and $A \in \mathcal{A}(\mathcal{T})$, there is a $\mathcal{T}$-safe forcing introducing an uncountable $B \subseteq A$ consisting of elements which pairwise conflict on C.

First we need another fact about suitable trees (see [15] for a proof).

Lemma 3.7. *Suppose T is coherent and every uncountable subset of T contains an uncountable antichain. Then for every $1 \leq n < \omega$ and uncountable $A \subseteq \bigotimes_{i<n} T$, there is an uncountable $B \subseteq A$ such that for all distinct $a, b \in B$, every element of a is incomparable with every element of b, and for every $i, j < n$, $\Delta(a(i), b(i)) = \Delta(a(j), b(j))$.*

Theorem 3.8. *Assume $\Xi(\mathcal{T})$. For every club C and every uncountable $A \in \mathcal{A}(\mathcal{T})$, there is a $\mathcal{T}$-safe forcing $\mathbb{Q}$ of size ω_1, and family $\{D_\xi : \xi < \omega_1\}$ of dense subsets of $\mathbb{Q}$, such that whenever G is filter of $\mathbb{Q}$ meeting every D_ξ, $\bigcup G$ is an uncountable subset of A of elements which pairwise conflict on C.*

Proof. Fix A and C as in the statement. Then for some $1 \leq n < \omega$, $A \subseteq \bigotimes_{i<n} T^i$ for some distinct $T^0, \ldots, T^{n-1} \in \mathcal{T}$. By going to an uncountable subset of A we may assume that A has at most one element for any given height $\alpha < \omega_1$. We will let $\mathbb{Q}$ be the set of finite subsets of A of elements which pairwise conflict on C ordered by reverse inclusion. We want to show that $\mathbb{Q}$ is $\mathcal{T}$-safe, so fix a R as in Theorem 3.1 and club E. We may assume that for all $(q, a), (q', a') \in R$, if $q \neq q'$, then $a \neq a'$. Towards a contradiction suppose that for no $(q, a), (q', a') \in R$, $q \not\perp q'$ and a conflicts with a' on E. We have that for some $1 \leq n' < \omega$ and

distinct $S^0,\dots,S^{n'-1}\in\mathcal{T}$, $\mathrm{ran}(R)$ is an uncountable subset of $\bigotimes_{i<n'}S^i$. The rest of the proof gets significantly simpler if we assume that both lists, $T^0,\dots,T^{n-1}$ and $S^0,\dots,S^{n'-1}$ are equal, so let us see why this reduction is possible first.

The easier part is to see that we can assume that $\{T^0,\dots,T^{n-1}\}\subseteq\{S^0,\dots,S^{n'-1}\}$, if not simply change R to R' where for each $(q,a)\in R$, $(q,a')\in R'$ is such that a' is a tuple of $\bigotimes\{T:T\in\{T^0,\dots,T^{n-1},S^0,\dots,S^{n'-1}\}\}$ which has a as a subtuple. Then we have that if a' and b' conflict on E, then so do a and b. Because of this we can assume that we only have one list $T^0,\dots,T^{n'-1}$, and that A is a subset of $\bigotimes_{i<n}T^i$ for some $n\leq n'$. Via a typical thinning out argument and Δ-system lemma, we may also assume that for $(q,a),(q',a')\in R$, if $q\neq q'$, then $q\cap q'=\emptyset$ and $|q|=|q'|=k$. Now pick a sufficiently large regular cardinal θ, and a club $F\subseteq\{N\cap\omega_1:|N|=\aleph_0,T^0,\dots,T^{n'-1},A,R,C,E\in N\prec H(\theta)\}$. For each $\nu\in F$, pick $(q_\nu,a_\nu)\in R$ such that every element in $q_\nu\cup\{a_\nu\}$ has height in $[\nu,\min(F\setminus(\nu+1))[$. Each q_ν is a finite set of elements of $\bigotimes_{i<n}T^i$ which pairwise conflict on C. Using elementarity one easily extends each tuple to one in $\bigotimes_{i<n'}T^i$ so that the resulting tuples remain pairwise conflicting on C.

Finally, this allows to assume that A and $\mathrm{ran}(R)$ are both subsets of $\bigotimes_{i<n}T^i$ for some $1\leq n<\omega$ and distinct $T^0,\dots,T^{n-1}\in\mathcal{T}$, and also the following: For each $(q,a),(q',a')\in R$, if $(q,a)\neq(q',a')$ then $q\cap q'=\emptyset$, $|q|=|q'|=:k$, and $a\neq a'$.

Again let F be a club as defined above, and let $\langle(q_\xi,a_\xi):\xi\in F\rangle$ be such that for each ξ, every element of $q_\xi\cup\{a_\xi\}$ has height in $[\xi,\min(F\setminus(\xi+1))[$. For each ξ, let $q_\xi=\{a_{\xi,0},\dots,a_{\xi,k-1}\}$ and $a_{\xi,k}=a_\xi$. We will be using the fact that $F\subseteq C\cap E$ through out the proof.

For each $\xi\in F$ and $i<n$ let $a^*_{\xi,i}:=(a_{\xi,0}(i),\dots,a_{\xi,k}(i))\restriction\xi$. Observe then that for all $j,j'\leq k$, $\Delta(a_{\xi,j}(i),a_{\xi,j'}(i))<\xi$ iff $a^*_{\xi,i}(j)\neq a^*_{\xi,i}(j')$. By pressing down we obtain a stationary $\Gamma\subseteq F$, $\gamma\in F$ with $\gamma<\min(\Gamma)$, $a^*_0,\dots,a^*_{n-1}$ such that:

(a) For every $i<n$, $a^*_i(0),\dots,a^*_i(k)\in T^i_\gamma$ (not necessarily distinct).
(b) For every $i<n$, for every $\xi\in\Gamma$, $a^*_i=a^*_{\xi,i}\restriction\gamma$.
(c) For every $i<n$, for every $j,j'\leq k$, either $a^*_i(j)\neq a^*_i(j')$, or $\Delta(a_{\xi,j}(i),a_{\xi,j'}(i))\geq\xi$ for all $\xi\in\Gamma$.

Applying Theorem 3.7 n-times we find uncountable $\Gamma'\subseteq\Gamma$ satisfying:

(d) For all $i<n$, for all $\xi<\eta$ in Γ', every element of $a^*_{\xi,i}$ is incomparable with every element of $a^*_{\eta,i}$ and for all $j<j'\leq k$, $\Delta(a^*_{\xi,i}(j),a^*_{\eta,i}(j))=\Delta(a^*_{\xi,i}(j'),a^*_{\eta,i}(j'))$.
(e) For all $\xi<\eta$ in Γ', every element appearing in $q_\xi\cup\{a_\xi\}$ has height less than η.

Finally an application of $\Xi(\mathcal{T})$ to the club F and the family $\{(a^*_{\xi,0}(0),\dots,a^*_{\xi,n-1}(0)):\xi\in\Gamma'\}$, gives us $\xi<\eta$ in Γ' satisfying that for all $i<i'<n$, $\nu_F(\Delta(a^*_{\xi,i}(0),a^*_{\eta,i}(0)))\neq\nu_F(\Delta(a^*_{\xi,i'}(0),a^*_{\eta,i'}(0)))$. However, by (d) we obtain the stronger:

(f) For every $i<i'<n$ and $j,j'\leq k$, $\nu_F(\Delta(a^*_{\xi,i}(j),a^*_{\eta,i}(j)))\neq\nu_F(\Delta(a^*_{\xi,i'}(j'),a^*_{\eta,i'}(j')))$.

We claim that q_ξ and q_η are compatible, and that a_ξ and a_η conflict on E. For this we need to do two things:

- Show that for all $j,j'<k$, $a_{\xi,j}$ and $a_{\eta,j'}$ are conflicting on C.
- Show that a_ξ and a_η conflict on E.

The second one follows directly from (a)–(f) and the fact that $F\subseteq E$. For the first one, the same argument, using $F\subseteq C$ works if $j=j'$. If $j\neq j'$ there are a few cases. In only one of these cases we actually use the fact that the elements of q_ξ (and q_η) pairwise conflict on C. Fix $i<i'<n$, we need to show that $\nu_C(\Delta(a_{\xi,j}(i),a_{\eta,j'}(i)))\neq\nu_C(\Delta(a_{\xi,j}(i'),a_{\eta,j'}(i')))$.

- Case $\xi \leq \Delta(a_{\xi,j}(i), a_{\xi,j'}(i))$ and $\xi \leq \Delta(a_{\xi,j}(i'), a_{\xi,j'}(i'))$. First observe that in this case we also have that $\eta \leq \Delta(a_{\eta,j}(i), a_{\eta,j'}(i)), \Delta(a_{\eta,j}(i'), a_{\eta,j'}(i'))$. So $\Delta(a_{\xi,j}(i), a_{\eta,j'}(i)) = \Delta(a^*_{\xi,i}(j), a^*_{\eta,i}(j))$ and $\Delta(a_{\xi,j}(i'), a_{\eta,j'}(i')) = \Delta(a^*_{\xi,i'}(j), a^*_{\eta,i'}(j))$ so we can conclude from (f).
- Case $\Delta(a_{\xi,j}(i), a_{\xi,j'}(i)) < \xi$ and $\xi \leq \Delta(a_{\xi,j}(i'), a_{\xi,j'}(i'))$. In this case we have that $\Delta(a_{\xi,j}(i), a_{\eta,j'}(i)) = \Delta(a^*_i(j), a^*_i(j')) < \gamma$ and $\gamma \leq \Delta(a_{\xi,j}(i'), a_{\eta,j'}(i')) = \Delta(a^*_{\xi,i'}(j), a^*_{\eta,i'}(j'))$ hold. But γ was chosen in C, so we are done.
- Case $\xi \leq \Delta(a_{\xi,j}(i), a_{\xi,j'}(i))$ and $\Delta(a_{\xi,j}(i'), a_{\xi,j'}(i')) < \xi$. This is symmetric to the previous case.
- Case $\Delta(a_{\xi,j}(i), a_{\xi,j'}(i)) < \xi$ and $\Delta(a_{\xi,j}(i'), a_{\xi,j'}(i')) < \xi$. Now we have that $\alpha_0 := \Delta(a_{\xi,j}(i), a_{\eta,j'}(i)) = \Delta(a_{\xi,j}(i), a_{\xi,j'}(i))$ and $\alpha_1 := \Delta(a_{\xi,j}(i'), a_{\eta,j'}(i')) = \Delta(a_{\xi,j}(i'), a_{\xi,j'}(i'))$. But $a_{\xi,j}, a_{\xi,j'} \in q_\xi$, thus they conflict on C, so $\nu_C(\alpha_0) \neq \nu_C(\alpha_1)$ and we are done.

We now know that $\mathbb{Q}$ is $\mathcal{T}$-safe. We need to construct our family of ω_1 many dense sets which will give us our uncountable subfamily of A of pairwise conflicting elements. One would hope that for each $\gamma < \omega_1$, $\{q \in \mathbb{Q} : (\exists a \in q)(\mathrm{ht}(a) > \gamma)\}$ is dense. But this is not necessarily the case.

For each $\alpha < \omega_1$ let $\mathbb{Q}^\alpha := \{q \in \mathbb{Q} : (\forall a \in q)(\alpha \leq \mathrm{ht}(a))\}$. Because $\mathbb{Q}^\alpha \subseteq \mathbb{Q}$, it should be clear that $\mathbb{Q}^\alpha$ is $\mathcal{T}$-safe. Thus to finish the proof of the theorem is enough to show that there is $\alpha < \omega_1$ such that for all $\gamma < \omega_1$, $\{q \in \mathbb{Q}^\alpha : (\exists a \in q)(\mathrm{ht}(a) > \gamma)\}$ is dense in $\mathbb{Q}^\alpha$. To see this, assume the contrary and construct a sequence $\langle \beta_\xi : \xi < \omega_1 \rangle$ and $\langle q_\xi : \xi < \omega_1 \rangle$ such that for all $\xi < \eta$,

- $\beta_\xi \leq \min\{\mathrm{ht}(a) : a \in q_\xi\}, \max\{\mathrm{ht}(a) : a \in q_\xi\} < \beta_\eta$.
- For no $a \in A$ with $\beta_\eta \leq \mathrm{ht}(a)$, $q_\xi \cup \{a\} \in \mathbb{Q}$.

Then $\{q_\xi : \xi < \omega_1\}$ is an uncountable antichain of $\mathbb{Q}$, a contradiction since we proved that $\mathbb{Q}$ is $\mathcal{T}$-safe, and thus ccc. □

4. Consistency results

The following is the analogous of Abraham-Shelah's [1, 6.15].

Theorem 4.1. *Let $\mathcal{T}$ be a family of suitable trees. Assume $\Xi(\mathcal{T})$ and $2^{\omega_1} = \omega_2$. Then there is a ccc forcing $\mathbb{M}(\mathcal{T})$ of size ω_2 such that $\mathsf{MA} + \Xi^+(\mathcal{T}) + 2^\omega = 2^{\omega_1} = \omega_2$ holds in any extension by $\mathbb{M}(\mathcal{T})$.*

Proof. This is just like the proof of $\mathsf{MA} + 2^\omega = 2^{\omega_1} = \omega_2$, but we just force with $\mathcal{T}$-safe forcing posets. Then, by Theorems 3.2 and 3.5 one obtains a model of MA_{ω_1} restricted to safe posets. By Theorem 3.8, $\Xi^+(\mathcal{T})$ holds in this model, and thus Theorem 3.4 implies that every ccc forcing is actually $\mathcal{T}$-safe, therefore the full MA_{ω_1} holds. □

Now, this implies that to finish the proof of Theorem A it is enough to show the consistency of "$2^{\omega_1} = \omega_2$ + there is a family $\mathcal{T}$ of 2^{ω_1}-many suitable trees satisfying $\Xi(\mathcal{T})$". The rest of this section is devoted to show that the existence of such a family follows from $\diamondsuit$, which of course is consistent with $2^{\omega_1} = \omega_2$. We will adapt the construction of 2^{ω_1}-many pairwise non club-isomorphic subtrees from [1] and [5]. A similar construction is also done in [3].

Let us denote by $\mathbb{S}$ the subtree of $\omega^{<\omega_1}$ of finite-to-one sequences. For $\alpha \leq \omega_1$, by a *uniformly coherent subtree of* $\mathbb{S} \restriction \alpha$, we mean a coherent subset $T \subseteq \mathbb{S} \restriction \alpha$ that is closed under taking initial segments and finite modifications, and has height α. We say that $b : \alpha \to \omega$ is *good for* T, if $b \in \mathbb{S}_\alpha$ and for all $\xi < \alpha$, $b \restriction \xi \in T$. Then is easy to see that there is a unique way of extending T to a uniformly coherent subtree of $\mathbb{S} \restriction (\alpha + 1)$ that contains b. We call this the *extension of T given by adjoining b.*

For $n \in \omega$ and $x, y \in \omega^{<\omega_1}$. we let $x \sqsubseteq_n y$ denote that $x \sqsubseteq y$ and for all $\xi \in \mathrm{dom}(y) \setminus \mathrm{dom}(x)$, $y(\xi) \geq n$. The following is easy to verify, and will be our main tool to extend uniformly coherent trees.

Lemma 4.2. *Let α be a countable limit ordinal. Let T be a uniformly coherent subtree of $\mathbb{S} \restriction \alpha$. Suppose that $\{t_n : n < \omega\} \subseteq T$ is such that for all $n < \omega$, $t_n \sqsubseteq_n t_{n+1}$ and $\sup\{\mathrm{dom}(t_n) : n < \omega\} = \alpha$. Then $b := \bigcup_{n<\omega} t_n$ is good for T.*

By a *finite modifier* we mean a partial finite function $m : \omega_1 \to \omega$. Given $t \in \mathbb{S}$ and a finite modifier m, $mt : \mathrm{dom}(t) \to \omega$ is defined by letting $mt(\xi) = t(\xi)$ if $\xi \notin \mathrm{dom}(m)$, and $mt(\xi) = m(\xi)$ otherwise.

Theorem 4.3. *Let α be a countable limit ordinal, n be a positive integer, $T^0, \dots, T^{n-1}$ be uniformly coherent subtrees of $\mathbb{S} \restriction \alpha$, $C \subseteq \alpha$ club, and $A \subseteq \bigotimes_{i<n} T^i$ be maximal with respect to having no two elements conflicting on C. Then there are $b_0, \dots, b_{n-1} \in \mathbb{S}_\alpha$ such that for all $i < n$, b_i is good for T^i, and whenever $b_0' =^* b_0, \cdots, b_{n-1}' =^* b_{n-1}$, some member of A conflicts on C with $(b_0', \dots, b_{n-1}')$.*

Proof. First note that we may assume that $\min(C) = 0$. Now fix $c \subseteq C$ cofinal in α, of order type ω satisfying that $0 \in c$, and for any two consecutive elements of c, there are at least $n-1$ element of C strictly between them. For $\xi \in c$ we let ξ^+ denote its successor in c, i.e., $\min(c \setminus (\xi + 1))$.

Let $\langle (m_{0,\xi}, \dots, m_{n-1,\xi}) : \xi \in c \rangle$ enumerate all n-element lists of finite modifiers with domain contained in α, allow repetitions so that each element appears infinitely many times. We may also assume that $\mathrm{dom}(m_{i,\xi}) \subseteq \xi$ for all $\xi \in c$ and $i < n$.

For each $i < n$, we will let $b_i := \bigcup_{\xi \in c} b_i(\xi)$, where for all $\xi \in c$, $b_i(\xi) \in T^i_\xi$ and $b_i(\xi) \sqsubseteq_{|c \cap \xi|} b_i(\xi^+)$. By Theorem 4.2, this will imply that each b_i will indeed be good for T^i. We construct $b_i(\xi)$ simultaneously for all i by recursion on ξ. Thus we must let $b_i(0) = \emptyset$ for all $i < n$. Suppose now that for some $\xi \in c$ we have already defined $b_i(\xi)$ for all $i < n$. Let $k := |c \cap \xi|$. Now we explain how to define the $b_i(\xi^+)$.

We claim that there exists $(x_0, \dots, x_{n-1}) \in \bigotimes_{i<n} T^i_{\xi^+}$ such that for all $i < n$, $b_i(\xi) \sqsubseteq_k x_i$, and $(m_{0,\xi} x_0, \dots, m_{n-1,\xi} x_{n-1}) \notin A$. Suppose not. By assumption on c, there are $\xi = \eta_0 < \cdots < \eta_n = \xi^+$ all in C. Using the uniform coherence we easily find $(x_0, \dots, x_{n-1}), (y_0, \dots, y_{n-1}) \in \bigotimes_{i<n} T^i_{\xi^+}$ such that for all $i < n$, $b_i(\xi) \sqsubseteq_k x_i, y_i$, $y_i \neq x_i$, and $\eta_i \leq \Delta(x_i, y_i) < \eta_{i+1}$. Since for all $i < n$, $\mathrm{dom}(m_{i,\xi}) \subseteq \xi$, we have that $m_{i,\xi} y_i \neq m_{i,\xi} x_i$ and $\eta_i \leq \Delta(m_{i,\xi} x_i, m_{i,\xi} y_i) < \eta_{i+1}$. By assumption, both $(m_{0,\xi} x_0, \dots, m_{n-1,\xi} x_{n-1})$ and $(m_{0,\xi} y_0, \dots, m_{n-1,\xi} y_{n-1})$ are elements of A. But this is impossible since by construction $(m_{0,\xi} x_0, \dots, m_{n-1,\xi} x_{n-1})$ and $(m_{0,\xi} y_0, \dots, m_{n-1,\xi} y_{n-1})$ are conflicting on C.

Using the claim we define $(b_0(\xi^+), \dots, b_{n-1}(\xi^+))$ as any $(x_0, \dots, x_{n-1}) \in \bigotimes_{i<n} T^i_{\xi^+}$ such that for all $i < n$, $b_i(\xi) \sqsubseteq_k x_i$, and $(m_{0,\xi} x_0, \dots, m_{n-1,\xi} x_{n-1}) \notin A$. This finishes the construction.

Let $b_0', \dots, b_{n-1}'$ be as in the statement of the lemma. Then there are finite modifiers $m_0, \dots, m_{n-1}$, such that for all $i < n$, $\mathrm{dom}(m_i) \subseteq \alpha$ and $b_i' = m_i b_i$. We need to show that $(m_0 b_0, \dots, m_{n-1} b_{n-1})$ conflicts on C with an element of A. Let $\xi \in c$ be such that $(m_0, \dots, m_{n-1}) = (m_{0,\xi}, \dots, m_{n-1,\xi})$. Then, by our construction, $(b_0(\xi^+), \dots, b_{n-1}(\xi^+))$ was chosen so that $(m_{0,\xi} b_0(\xi^+), \dots, m_{n-1,\xi} b_{n-1}(\xi^+))$ is not in A. Then, the maximality of A implies that there must be some $a \in A$ conflicting on C with $(m_{0,\xi} b_0(\xi^+), \dots, m_{n-1,\xi} b_{n-1}(\xi^+))$. Clearly $(b_0', \dots, b_{n-1}') = (m_0 b_0, \dots, m_{n-1} b_{n-1})$ extends $(m_{0,\xi} b_0(\xi^+), \dots, m_{n-1,\xi} b_{n-1}(\xi^+))$ so we conclude that $(b_0', \dots, b_{n-1}')$ also conflicts on C with a. □

Via a typical coding and decoding argument we obtain the following variation of $\diamondsuit$. (see [2] for a more general version).

Lemma 4.4. *Assume $\diamondsuit$. There is a sequence $\langle X_\alpha : \alpha < \omega_1\rangle$ of elements of $H(\omega_1)$ such that whenever $1 \leq n < \omega$, $t_0, \dots, t_{n-1} \in 2^{\omega_1}$, $C \subseteq \omega_1$, and $A \subseteq \bigotimes_{i<n} \mathbb{S}$ is such that for all $\alpha < \omega_1$, $A \restriction \alpha := \{a \in A : \mathrm{ht}(a) < \alpha\}$ is countable, there are stationarily many $\alpha < \omega_1$ such that*

- *$X_\alpha = (t_{\alpha,0}, \dots, t_{\alpha,n-1}, c_\alpha, A_\alpha) \in (2^\alpha)^n \times [\alpha]^{\leq\omega} \times [\bigotimes_{i<n} \mathbb{S} \restriction \alpha]^{\leq\omega}$.*
- *$C \cap \alpha = c_\alpha$.*
- *For all $i < n$, $t_i \restriction \alpha = t_{\alpha,i}$.*
- *$A \restriction \alpha = A_\alpha$.*

Moreover, replacing some of the X_α with dummy values we may assume that all of the X_α are elements of $(2^\alpha)^n \times [\alpha]^{\leq\omega} \times [\bigotimes_{i<n} \mathbb{S} \restriction \alpha]^{\leq\omega}$ for some $1 \leq n < \omega$.

Theorem 4.5. *Assume $\diamondsuit$. There is a family $\mathcal{T}$ of uniformly coherent subtrees of $\mathbb{S}$ satisfying $\Xi(\mathcal{T})$ and $|\mathcal{T}| = 2^{\omega_1}$.*

Proof. We build a sequence $\langle T^s : s \in 2^{\leq\omega_1}\rangle$ such that for all $\alpha \leq \omega_1$ and $s \in 2^\alpha$, T^s is a uniformly coherent subtree of $\mathbb{S} \restriction \alpha$, and for all $\beta < \alpha$, $T^{s\restriction\beta} = T^s \restriction \beta$. We want $\{T^t : t \in 2^{\omega_1}\}$ to witness the theorem. Fix $\langle X_\alpha : \alpha < \omega_1\rangle$ a $\diamondsuit$-sequence as in Theorem 4.4.

Recursively for $\alpha < \omega_1$, we define T^t for all $t \in 2^\alpha$. Observe that we are forced to set $T^\emptyset = \emptyset$ and $T^{\langle 0\rangle} = T^{\langle 1\rangle} = \{\emptyset\}$. This handles $\alpha = 0$ and $\alpha = 1$. At limits we must take the union, i.e., if α is limit, then $T^t := \bigcup_{\xi<\alpha} T^{t\restriction\xi}$ for all $t \in 2^\alpha$. Now assume that for some $0 < \alpha$ we have defined T^t for all $t \in 2^\alpha$, we need to define $T^{t^\frown\langle 0\rangle}$ and $T^{t^\frown\langle 1\rangle}$ for all $t \in 2^\alpha$. If α is successor, then for every $t \in 2^\alpha$, T^t has a last level, and thus the uniform coherence says that both, $T^{t^\frown\langle 0\rangle}$ and $T^{t^\frown\langle 1\rangle}$, must be equal to $T^t \cup \{x^\frown\langle n\rangle : x \in T^t_{\alpha-1}, n < \omega\}$. It only remains to define $T^{t^\frown\langle 0\rangle}$ and $T^{t^\frown\langle 1\rangle}$ for $t \in 2^\alpha$ with limit α.

Let $1 \leq n < \omega$ be such that $X_\alpha = (t_{\alpha,0}, \dots, t_{\alpha,n-1}, c_\alpha, A_\alpha) \in (2^\alpha)^n \times [\alpha]^{\leq\omega} \times [\bigotimes_{i<n} \mathbb{S} \restriction \alpha]^{\leq\omega}$. For each $t \in 2^\alpha$, fix T'^t some uniformly coherent subtree of $\mathbb{S} \restriction (\alpha+1)$ extending T^t. This exists by Theorem 4.2. For each $t \in 2^\alpha$ not in $\{t_{\alpha,0}, \dots, t_{\alpha,n-1}\}$ define $T^{t^\frown\langle 0\rangle}$ and $T^{t^\frown\langle 1\rangle}$ both be equal to T'^t. If any of the following conditions fail, do the same for all the $t \in \{t_{\alpha,0}, \dots, t_{\alpha,n-1}\}$.

- For all $i < j < n$, $t_{\alpha,i} \neq t_{\alpha,j}$.
- c_α is unbounded in α.
- $A_\alpha \subseteq \bigotimes_i T^{t_{\alpha,i}}$ is such that no two elements of A_α conflict on c_α, and is maximal with respect to this property.

Otherwise, apply Theorem 4.3 to $T^{t_{\alpha,0}}, \dots, T^{t_{\alpha,n-1}}$, using $A = A_\alpha$ and $C = c_\alpha$, to obtain $b_0, \dots, b_{n-1}$. Then, for all $i < n$ define both $T^{t^\frown_{\alpha,i}\langle 0\rangle}$ and $T^{t^\frown_{\alpha,i}\langle 1\rangle}$ as the result of adjoining b_i to $T^{t_{\alpha,i}}$. This finishes the construction of our family.

Let $\mathcal{T} = \{T^t : t \in 2^{\omega_1}\}$. We claim that $\Xi(\mathcal{T})$ holds. For this fix $1 \leq n < \omega$ and $t_0, \dots, t_{n-1}$ distinct elements of 2^{ω_1}. Also fix C a club and $A \subseteq \bigotimes_{i<n} T^{t_i}$ be such that no two elements of A conflict on C. Extending A if necessary, we may assume that A is maximal with this property. We now show that A is countable.

Let $\gamma < \omega_1$ be such that for all $i < j < n$, $t_i \restriction \gamma \neq t_j \restriction \gamma$. One easily finds a club $D \subseteq C$ such that for all $\alpha \in D$, α is limit of elements in C and $A \restriction \alpha$ is a maximal subset of $\bigotimes_{i<n} T^{t_i} \restriction \alpha$ having no two elements conflicting on $C \cap \alpha$. We may also assume that $\gamma < \min(D)$. Using the diamond property, let $\alpha \in D$ be such that $C \cap \alpha = c_\alpha$, $t_i \restriction \alpha = t_{\alpha,i}$ for all $i < n$, and $A \restriction \alpha = A_\alpha$. Let $f : n \to 2$ be such that $t_i \restriction (\alpha+1) = t^\frown_{\alpha,i}\langle f(i)\rangle$.

We claim that $A = A_\alpha$. If not, there is $a \in A$ of height at least α. The maximality implies that in fact A must have an element of height exactly α, that is, an element $(b'_0, \dots, b'_{n-1})$ where each b'_i is an element of $T^{t_{\alpha,i}\widehat{\ }\langle f(i)\rangle}$ of height α. But by the choice of $b_0, \dots, b_{n-1}$ in the α-th step of the construction, this implies that $(b'_0, \dots, b'_{n-1})$ conflicts on C with an element of $A_\alpha \subseteq A$. So we conclude that $A = A_\alpha$. □

5. Final remarks and open questions

Is it possible to construct using $\diamondsuit$ (or stronger variations) a family $\mathcal{T}$ of 2^{ω_1}-many suitable trees satisfying the stronger Ξ^+? A positive answer would render the work of Section 3 unnecessary, since by Theorem 3.4, the standard forcing to obtain $\mathsf{MA} + \neg\mathsf{CH}$ would preserve $\Xi(\mathcal{T})$.

Observe that in our final model, we not only have $\Xi(\mathcal{T})$, but $\Xi^+(\mathcal{T})$. So no ccc forcing can make the trees in $\mathcal{T}$ club-isomorphic again. Does Ξ^+ have any other niceties over Ξ?

Although we used $\diamondsuit$ to obtain the consistency of "$2^{\omega_1} = \omega_2$ + there is a family $\mathcal{T}$ of 2^{ω_1}-many suitable trees satisfying $\Xi(\mathcal{T})$", another option would be to iterate: The starting point comes from the observation that given a suitable family satisfying $\Xi(\mathcal{T})$, there is a $\mathcal{T}$-safe forcing extension in which there is a suitable $U \notin \mathcal{T}$ such that $\Xi(\mathcal{T} \cup \{U\})$. To see this, start from a family $\mathcal{T}$ satisfying $\Xi(\mathcal{T})$, we can first force with $\mathbb{M}(\mathcal{T})$ so that it also holds $\Xi^+(\mathcal{T})$ and that every tree in $\mathcal{T}$ is special, and then add a single Cohen real, which being a countable forcing, preserves both $\Xi^+(\mathcal{T})$ and that every tree in $\mathcal{T}$ is special. In the resulting model there is a suitable tree without stationary antichains (see [15] and [4]). Now we use the following.

Proposition 5.1. *Assume $\Xi^+(\mathcal{T})$ and that every $T \in \mathcal{T}$ is special. Suppose that U is a suitable tree with no stationary antichains, then $\Xi(\mathcal{T} \cup \{U\})$.*

Proof. The result is trivial if $\mathcal{T}$ is empty. Fix $1 \leq n < \omega$, distinct $T^0, \dots, T^{n-1} \in \mathcal{T}$, a club C, and uncountable $A \subseteq T^0 \otimes \cdots \otimes T^{n-1} \otimes U$. We need to show that there are $a, b \in A$ conflicting on C. By $\Xi^+(\mathcal{T})$ and the fact that U is suitable, going to an uncountable subset we may assume that the elements of $\{(a(0), \dots, a(n-1)) : a \in A\}$ are pairwise conflicting on C and that $\{a(n) : a \in A\}$ is an antichain.

Let $\{a_\xi : \xi < \omega_1\}$ enumerate an uncountable subset of A such that for each $\xi < \omega_1$, $\xi \leq \mathrm{ht}(a_\xi)$. For each $\xi < \omega_1$, let $b_\xi := (a_\xi(0) \restriction \xi, \dots, a_\xi(n) \restriction \xi)$. Because for each $i < n$, T^i is special, we find stationary $\Gamma \subseteq C$ such that for all $\xi < \eta$ and for all $i < n$, $b_\xi(i)$ is incomparable with $b_\eta(i)$. Now because U has no stationary antichains, we find $\xi < \eta$ in Γ such that $b_\xi(n) \leq_T b_\eta(n)$. Observe then that for all $i < n$, $\Delta(a_\xi(i), a_\eta(i)) < \xi \leq \Delta(a_\xi(n), a_\eta(n))$. Since $\xi \in C$, we conclude that a_ξ and a_η conflict on C. □

Of course, to have the family have size 2^{ω_1} one has to make an iteration of length ω_2 preserving $2^{\omega_1} = \omega_2$, which requires more care.

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